MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY TEST – 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions — MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If $5\tan\beta = 4$, then $\dfrac{5\sin\beta - 2\cos\beta}{5\sin\beta + 2\cos\beta}$ is equal to: [1]

(A) $\frac{1}{3}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) 6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{1}{3}$

Justification: Dividing numerator and denominator by $\cos\beta$: $\dfrac{5\tan\beta - 2}{5\tan\beta + 2} = \dfrac{4-2}{4+2} = \dfrac{2}{6} = \dfrac{1}{3}$.

Q2. The value of $\dfrac{2\tan 30^\circ}{1+\tan^2 30^\circ}$ is equal to: [1]

(A) $\sin 60^\circ$ (B) $\cos 60^\circ$
(C) $\tan 60^\circ$ (D) $\sin 30^\circ$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\sin 60^\circ$

Justification: $\dfrac{2(1/\sqrt{3})}{1+1/3} = \dfrac{2/\sqrt{3}}{4/3} = \dfrac{3}{2\sqrt{3}} = \dfrac{\sqrt{3}}{2} = \sin 60^\circ$.

Q3. The value of $\dfrac{1-\tan^2 45^\circ}{1+\tan^2 45^\circ}$ is equal to: [1]

(A) $\tan 90^\circ$ (B) 1 (C) $\sin 45^\circ$ (D) 0

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) 0

Justification: Since $\tan 45^\circ = 1$, $\dfrac{1-1^2}{1+1^2} = \dfrac{0}{2} = 0$.

Q4. If $\sin 2A = 2\sin A$ is true, then the value of A is: [1]

(A) 0° (B) 30° (C) 45° (D) 60°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 0°

Justification: At $A = 0^\circ$: LHS $= \sin 0^\circ = 0$ and RHS $= 2\sin 0^\circ = 0$. For $30^\circ, 45^\circ, 60^\circ$, LHS $\neq$ RHS.

Q5. **Assertion (A):** The value of $\cos\theta$ increases as θ increases from 0° to 90°.
Reason (R): $\cos 0^\circ = 1$ and $\cos 90^\circ = 0$, so $\cos\theta$ decreases as θ increases. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

Justification: As θ goes from 0° to 90°, $\cos\theta$ decreases from 1 to 0. Thus (A) is false and (R) is true.

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Q6. If $\sin(A-B)=\frac{1}{2}$ and $\cos(A+B)=\frac{1}{2}$, where $0^\circ < A+B \le 90^\circ$ and $A > B$, find the values of acute angles A and B. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Form Linear Equations in A and B [1 Mark]

$$\sin(A-B)=\frac{1}{2}=\sin 30^\circ \implies A-B=30^\circ \quad \text{— (i)}$$

$$\cos(A+B)=\frac{1}{2}=\cos 60^\circ \implies A+B=60^\circ \quad \text{— (ii)}$$

Step 2: Solve for A and B [1 Mark]

Adding (i) and (ii): $2A=90^\circ \implies A=45^\circ$. Substituting into (ii): $B=60^\circ-45^\circ=15^\circ$.

$$\boxed{A=\mathbf{45^\circ},\qquad B=\mathbf{15^\circ}}$$

Q7. Prove that: $\dfrac{\tan\theta}{1-\cot\theta}+\dfrac{\cot\theta}{1-\tan\theta}=1+\sec\theta\csc\theta$. [3]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Convert into $\sin\theta$ and $\cos\theta$ [1 Mark]

$$\text{LHS}=\frac{\sin\theta/\cos\theta}{1-\cos\theta/\sin\theta}+\frac{\cos\theta/\sin\theta}{1-\sin\theta/\cos\theta}=\frac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)}-\frac{\cos^2\theta}{\sin\theta(\sin\theta-\cos\theta)}$$

Step 2: Combine Terms & Apply $a^3-b^3=(a-b)(a^2+ab+b^2)$ [1.5 Marks]

$$\text{LHS}=\frac{\sin^3\theta-\cos^3\theta}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}=\frac{(\sin\theta-\cos\theta)(\sin^2\theta+\sin\theta\cos\theta+\cos^2\theta)}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}$$

Step 3: Separate Terms to Obtain RHS [0.5 Mark]

$$\text{LHS}=\frac{1+\sin\theta\cos\theta}{\sin\theta\cos\theta}=\frac{1}{\sin\theta\cos\theta}+1=\boxed{1+\sec\theta\csc\theta=\text{RHS}\quad[\text{Hence Proved}]}$$

Q8. Prove that: $(\csc A - \sin A)(\sec A - \cos A) = \dfrac{1}{\tan A + \cot A}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Simplify LHS [1.5 Marks]

$$\text{LHS} = \left(\frac{1}{\sin A} - \sin A\right)\left(\frac{1}{\cos A} - \cos A\right) = \left(\frac{1-\sin^2 A}{\sin A}\right)\left(\frac{1-\cos^2 A}{\cos A}\right)$$

$$= \frac{\cos^2 A}{\sin A} \cdot \frac{\sin^2 A}{\cos A} = \sin A \cos A$$

Step 2: Simplify RHS [1.5 Marks]

$$\text{RHS} = \frac{1}{\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}} = \frac{\sin A \cos A}{\sin^2 A + \cos^2 A} = \frac{\sin A \cos A}{1} = \sin A \cos A$$

$$\boxed{\therefore \text{LHS} = \text{RHS} = \sin \mathbf{A} \cos \mathbf{A} \quad \text{[Hence Proved]}}$$

Q9. Prove that: $\left(\dfrac{1+\tan^2 A}{1+\cot^2 A}\right) = \left(\dfrac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Simplify First Expression [1.5 Marks]

$$\frac{1+\tan^2 A}{1+\cot^2 A} = \frac{\sec^2 A}{\csc^2 A} = \frac{1/\cos^2 A}{1/\sin^2 A} = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A$$

Step 2: Simplify Second Expression [1.5 Marks]

$$\left(\frac{1-\tan A}{1-\cot A}\right)^2 = \left(\frac{1-\tan A}{1-\frac{1}{\tan A}}\right)^2 = \left(\frac{1-\tan A}{\frac{\tan A - 1}{\tan A}}\right)^2 = (-\tan A)^2 = \tan^2 A$$

$$\boxed{\therefore \frac{1+\tan^2 A}{1+\cot^2 A} = \left(\frac{1-\tan A}{1-\cot A}\right)^2 = \tan^{\mathbf{2}} \mathbf{A} \quad \text{[Hence Proved]}}$$

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Q10. Maritime Kite Propulsion (Skysails Engineering)

A modern cargo ship uses an auxiliary towing kite system ('Skysails') to reduce fuel consumption. The towing kite flies high above the ocean surface, attached to the bow by a high-strength tether cable OT of length 150 m. The cable makes an angle of $\theta = 45^\circ$ with the horizontal deck of the ship. **Based on the above geometric model, answer the following questions:**

(i) Find the vertical operating altitude h of the towing kite above the deck. [1]

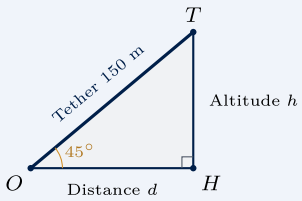
(ii) Find the horizontal forward pulling distance d from the attachment point to the kite's vertical plumb line. [1]

(iii) If the total tension force along the tether cable is $T = 80$ kN, calculate the horizontal forward propulsive force component $F_x = T\cos 45^\circ$ and vertical lift force $F_y = T\sin 45^\circ$. [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Vertical Operating Altitude h [1 Mark]

$$h = OT\sin 45^\circ = 150 \times \frac{1}{\sqrt{2}} = \boxed{\mathbf{75\sqrt{2}}\text{ m} \approx 106.07\text{ m}}$$



Sub-part (ii): Horizontal Distance d [1 Mark]

$$d = OT\cos 45^\circ = 150 \times \frac{1}{\sqrt{2}} = \boxed{\mathbf{75\sqrt{2}}\text{ m} \approx 106.07\text{ m}}$$

Sub-part (iii): Propulsive & Lift Force Components [2 Marks]

$$F_x = 80\cos 45^\circ = \frac{80}{\sqrt{2}} = \mathbf{40\sqrt{2}}\text{ kN} \approx 56.57\text{ kN}$$

$$F_y = 80\sin 45^\circ = \frac{80}{\sqrt{2}} = \mathbf{40\sqrt{2}}\text{ kN} \approx 56.57\text{ kN}$$

$$\boxed{F_x = F_y = \mathbf{40\sqrt{2}}\text{ kN} \approx \mathbf{56.57}\text{ kN}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★